

Student Registration System

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

PREPARED BY: MAHA DARWESH – BUSINESS ANALYST
DOCUMENT VERSION: 1.0
STATUS: DRAFT
DATE: AUGUST 2024

Contents

Document Control	1
1. Introduction	3
1.1 Purpose	3
1.2 Scope	3
1.3 Intended Audience	3
1.4 Definitions and Acronyms	3
2. System Overview	3
2.1 Product Perspective	3
2.2 High-Level Capabilities	3
2.3 User Roles	4
3. Assumptions, Constraints & Dependencies	4
4. Functional Requirements	4
5. Non-Functional Requirements	7
6. Use Case Requirements	7
7. Business Rules	9
8. Data Requirements	9
9. External Interface Requirements	10
10. User Interface Requirements	10
11. Security & Access Requirements	11
12. Error Handling, Logging & Monitoring	11
13. Reporting Requirements	11
14. Traceability	11
15. Acceptance & Verification	11
Appendix A – Recommended SRS Artifact References	11
Appendix B – Quality Checklist	12

Document Control

Field	Value
-------	-------

Document Title	Student Registration System – Software Requirements Specification
Document ID	SRS-SRS-001
Version	1.0
Status	Draft
Document Type	Software Requirements Specification
Intended Audience	Business stakeholders, Product Owner, Business Analysts, Architects, Developers, QA/UAT teams
Classification	Business Analysis Portfolio / Project Documentation

1. Introduction

1.1 Purpose

This Software Requirements Specification defines the system-level functional and non-functional requirements for the Student Registration System. It provides a central specification and traceability point for detailed artifacts such as Use Case Specifications, BPMN models, Business Rules, Wireframes, the Data Model, and the Requirements Traceability Matrix (RTM).

1.2 Scope

The system supports authentication and authorization, course search, registration and enrollment, academic eligibility validation, seat allocation and waitlisting, schedule synchronization, tuition and invoicing, financial aid, online payments, notifications, reporting, administrative configuration, auditability, and enterprise integrations.

1.3 Intended Audience

Business stakeholders, Registrar and Academic Operations, Finance/Bursar teams, Financial Aid teams, Product Owner, Business Analysts, Solution/Software Architects, Developers, QA engineers, and UAT participants.

1.4 Definitions and Acronyms

Term	Meaning
SRS	Software Requirements Specification
FR	Functional Requirement
NFR	Non-Functional Requirement
UC	Use Case
BPMN	Business Process Model and Notation
RTM	Requirements Traceability Matrix
UAT	User Acceptance Testing
MFA	Multi-Factor Authentication
RBAC	Role-Based Access Control
SIS	Student Information System

2. System Overview

2.1 Product Perspective

The Student Registration System operates as an integrated university application that coordinates student registration activities with identity services, student/academic data, financial services, notification services, and audit/monitoring capabilities.

2.2 High-Level Capabilities

Secure authentication and role-based access
Course and section search
Registration eligibility validation
Seat allocation, concurrency control and waitlist management
Add/drop/withdraw processing
Schedule and registration-status synchronization

Tuition, invoice and financial-aid processing

Online payment and reconciliation

Notifications and communications

Reporting, audit and administration

2.3 User Roles

Role	Primary Responsibilities
Student	Search courses, register, manage schedule, review charges, make payments, receive notifications.
Registrar Staff	Support registration operations, manage approved exceptions and student registration status.
Academic Advisor	Review academic eligibility and selected exception/approval requests.
Finance / Bursar Staff	Manage invoices, charges, payment reconciliation and financial exceptions.
Financial Aid Staff	Maintain and review eligible aid/waiver information.
System Administrator	Manage configuration, roles, access and operational administration.

3. Assumptions, Constraints & Dependencies

Authoritative student and academic data are available from approved institutional source systems.

Identity and Access Management services are available for authentication, MFA and role/permission information.

Registration actions must comply with configured academic policies, registration windows and capacity rules.

Financial transactions must use approved payment and financial systems.

Detailed business rules are maintained in the Business Rules Catalog and referenced rather than duplicated in full.

4. Functional Requirements

The complete Functional Requirements Catalog contains FR-001 through FR-035. The examples below demonstrate the recommended SRS presentation style.

ID	Requirement	Module	Priority
FR-001	User Login	Authentication	High
FR-002	Forgot Password	Authentication	High
FR-003	Change Password	Authentication	High
FR-004	User Logout	Authentication	High
FR-005	Account Lockout	Authentication	High
FR-006	Session Timeout	Authentication	High
FR-007	Remember Me	Authentication	Medium

FR-008	Multi-Factor Authentication (MFA)	Authentication	High
FR-009	Concurrent Session Management	Authentication	High
FR-010	Role-Based Access Control (RBAC)	Authorization & Access Control	High
FR-011	Student Registration & Enrollment Management	Registration & Enrollment	High
FR-012	Course Search & Section Availability	Course Search & Registration	High
FR-013	Prerequisite & Co-requisite Validation	Academic Eligibility Validation	Critical
FR-014	Schedule Conflict Validation	Registration Controls	Critical
FR-015	Credit-Hour Limit Validation	Registration Controls	Critical
FR-016	Hold & Academic Eligibility Validation	Registration Controls	High
FR-017	Registration Window & Deadline Validation	Registration Controls	High
FR-018	Real-Time Seat Capacity & Concurrency Control	Registration Controls	High
FR-019	Duplicate Request Prevention & Idempotency	Registration Controls	High
FR-020	Add / Drop / Withdraw Management	Enrollment Lifecycle	High
FR-021	Waitlist Management	Enrollment Lifecycle	High
FR-022	Schedule & Registration Status Synchronization	Enrollment Lifecycle	High

FR-023	Tuition & Fee Calculation	Financial Management	High
FR-024	Invoice & Student Account Posting	Financial Management	High
FR-025	Online Payment Processing	Financial Management	High
FR-026	Scholarships, Waivers & Discounts	Financial Management	High
FR-027	Notifications & Communications	Notifications & Communications	High
FR-028	Exception & Approval Workflow Management	Governance	High
FR-029	Academic Policy & Rule Administration	Academic Operations	High
FR-030	Course Section & Offering Administration	Academic Operations	High
FR-031	Audit Trail & Compliance Monitoring	Governance & Assurance	High
FR-032	Reporting & Management Dashboards	Reporting & Analytics	Medium
FR-033	Student Profile & Academic Status Management	Student Administration	High
FR-034	Academic Term & Calendar Management	Academic Operations	High
FR-035	Enterprise Integration & Data Synchronization	Integration	High

5. Non-Functional Requirements

ID	Category	Requirement	Verification
NFR-SEC	Security	Sensitive authentication and session data shall be protected using approved security controls in transit and at rest where applicable.	Security verification / penetration testing
NFR-PERF	Performance	Course-search and registration transactions shall meet approved response-time targets under defined normal and peak-load conditions.	Performance/load testing
NFR-AVL	Availability	The system shall meet the approved service-availability target during published registration periods, excluding approved maintenance windows.	Availability monitoring
NFR-REL	Reliability	Confirmed registration and financial transactions shall not be lost or duplicated during recoverable service failures.	Recovery and transaction-integrity testing
NFR-CON	Concurrency	Concurrent requests for limited seats shall not cause confirmed enrollment to exceed configured section capacity.	Concurrency/load testing
NFR-AUD	Auditability	Security-sensitive, registration, approval and financial actions shall produce traceable audit records according to approved policy.	Audit verification
NFR-USA	Usability & Accessibility	User-facing functions shall follow approved usability and accessibility standards for the target user population.	Accessibility/usability review
NFR-INT	Integration	Enterprise interfaces shall detect, log and safely handle failed or duplicate integration messages.	Integration testing

6. Use Case Requirements

Detailed actor-system interaction steps are maintained in the Use Case Specification Master. The SRS references them to avoid duplicating Main Success Flows, Alternative Flows and Exception Flows.

Use Case	Name	Related FR
UC-001	Student Login & Authentication	FR-001, FR-005, FR-006, FR-008, FR-009, FR-010, FR-031
UC-002	Forgot Password / Password Recovery	FR-002, FR-005, FR-008, FR-027, FR-031
UC-003	Change Password	FR-003, FR-008, FR-010, FR-027, FR-031
UC-004	Secure User Logout	FR-004, FR-006, FR-007, FR-009, FR-031
UC-005	Session Timeout & Automatic Logout	FR-004, FR-006, FR-031
UC-006	Remember Me / Trusted Login	FR-001, FR-004, FR-005, FR-006, FR-007, FR-008, FR-031
UC-007	Multi-Factor / Step-Up Authentication	FR-001, FR-008, FR-010, FR-031
UC-008	Concurrent Session Management	FR-004, FR-006, FR-009, FR-031
UC-009	Role-Based Access Control (RBAC)	FR-008, FR-010, FR-031
UC-010	Student Course Registration	FR-010, FR-011, FR-012, FR-013, FR-014, FR-015, FR-016, FR-017, FR-018, FR-019, FR-027, FR-031, FR-033, FR-034
UC-011	Course Search & Section Availability	FR-012, FR-018, FR-030, FR-034
UC-012	Registration Eligibility Validation	FR-013, FR-014, FR-015, FR-016, FR-017, FR-029, FR-033, FR-034
UC-013	Seat Allocation & Concurrent Registration	FR-011, FR-018, FR-019, FR-031
UC-014	Add / Drop / Withdraw Course	FR-020, FR-022, FR-023, FR-024, FR-027, FR-034
UC-015	Waitlist Management	FR-018, FR-021, FR-027, FR-034
UC-016	Schedule & Registration Status Synchronization	FR-011, FR-020, FR-021, FR-022, FR-031, FR-035
UC-017	Tuition & Fee Calculation	FR-020, FR-023, FR-026, FR-029, FR-034
UC-018	Invoice & Student Account Posting	FR-023, FR-024, FR-026, FR-031, FR-035
UC-019	Online Payment Processing	FR-024, FR-025, FR-027, FR-031, FR-035
UC-020	Scholarships, Waivers & Discounts	FR-023, FR-024, FR-026, FR-028, FR-031
UC-021	Notifications & Communications	FR-027, FR-031, FR-035
UC-022	Exception & Approval Workflow	FR-008, FR-010, FR-028, FR-029, FR-031
UC-023	Academic Policy & Rule Administration	FR-010, FR-013, FR-014, FR-015, FR-016, FR-017, FR-028, FR-029, FR-031
UC-024	Course Section & Offering Administration	FR-010, FR-012, FR-018, FR-021, FR-027, FR-029, FR-030, FR-031, FR-034, FR-035
UC-025	Audit Trail & Compliance Monitoring	FR-001–FR-035; FR-031 primary
UC-026	Reporting & Management Dashboards	FR-010, FR-018, FR-021, FR-023, FR-026, FR-028, FR-029, FR-030, FR-031, FR-032, FR-033, FR-034, FR-035

UC-027	Student Profile & Academic Status Management	FR-010, FR-011, FR-016, FR-020, FR-023, FR-026, FR-028, FR-029, FR-030, FR-031, FR-032, FR-033, FR-035
UC-028	Academic Term & Calendar Management	FR-010, FR-011, FR-012, FR-017, FR-020, FR-021, FR-022, FR-023, FR-027, FR-029, FR-030, FR-031, FR-032, FR-034, FR-035
UC-029	Enterprise Integration & Data Synchronization	FR-019, FR-022, FR-024, FR-025, FR-027, FR-031, FR-033, FR-035

7. Business Rules

Business rules are maintained in the Business Rules Catalog. The SRS should reference rule IDs from individual requirements and the RTM rather than repeating full rule definitions.

Registration eligibility rules
 Prerequisite/co-requisite rules
 Credit-hour limits
 Registration-window/deadline rules
 Seat-capacity and waitlist rules
 Payment and financial-aid rules
 Authentication and security policies

8. Data Requirements

Detailed logical definitions are maintained in the Data Model & Data Dictionary. Core entities include:

Student
 User Account / Role
 Academic Term
 Course
 Course Section
 Enrollment
 Waitlist Entry
 Invoice / Invoice Line
 Payment / Payment Allocation
 Financial Aid Award
 Notification
 Audit Event

Key data principles include referential integrity, prevention of duplicate active enrollment, capacity consistency, authoritative-source ownership, traceable status changes, and secure handling of sensitive information.

9. External Interface Requirements

Interface	Purpose	Key Data / Interaction	Failure Handling
Identity & Access Management	Authentication, MFA, authorization	User identity, authentication result, roles/permissions	Deny unsafe access; log service/authentication failures.
Student Information System (SIS)	Authoritative student/academic information	Student status, program, academic attributes	Prevent inconsistent updates; retry/reconcile according to integration policy.
Academic Policy Service	Eligibility/rule evaluation	Prerequisites, holds, limits, rule outcomes	Return controlled validation outcome and reason.
Payment Gateway / Bank	Online payment authorization/settlement	Payment request, authorization result, settlement reference	Do not post success without authoritative confirmation; reconcile discrepancies.
Financial Aid System	Aid/waiver information	Approved award/waiver and applicability	Do not apply unapproved or invalid awards.
Notification Service	User communications	Event, recipient, template/channel	Track failed delivery and retry according to policy.
Audit / Monitoring Service	Audit and operational monitoring	Security/business events, correlation references	Preserve traceability and operational visibility.

10. User Interface Requirements

Detailed low-fidelity wireframes are maintained as separate portfolio artifacts. The SRS references the screen IDs and key functional expectations.

Wireframe	Screen	Primary Actor
WF-001	Secure Sign In	Student / Authorized User
WF-002	MFA / Step-Up Verification	Authorized User
WF-003	Student Dashboard	Student
WF-004	Course Search & Filters	Student
WF-005	Section Details & Availability	Student
WF-006	Registration Cart & Validation	Student
WF-007	Registration Result	Student
WF-008	My Schedule & Enrollment Status	Student
WF-009	Add / Drop / Withdraw Confirmation	Student
WF-010	Waitlist Status	Student
WF-011	Student Account & Invoice	Student
WF-012	Online Payment & Reconciliation	Student
WF-013	Notifications Center	Student / Authorized User
WF-014	Exception Request	Student / Authorized Requester
WF-015	Approval Work Queue	Approver
WF-016	Academic Policy Administration	Registrar / Policy Administrator
WF-017	Course Section Administration	Registrar / Department Staff
WF-018	Audit, Reports & Integration Health	Auditor / Operations / Manager

11. Security & Access Requirements

Authentication shall use approved institutional identity controls.

MFA shall be enforced when required by security policy.

RBAC shall restrict protected functions and data according to assigned permissions.

Session timeout and concurrent-session controls shall follow approved security policy.

Failed authentication and security-sensitive actions shall be auditable.

Sensitive information shall not be exposed in unauthorized user interfaces, logs, or integration responses.

12. Error Handling, Logging & Monitoring

User-facing errors shall be understandable and shall not disclose sensitive technical/security information.

Transactional failures shall not leave enrollment, seat, invoice, or payment data in an inconsistent state.

Important failures shall include traceable correlation/reference information for support and audit.

Operational monitoring shall cover critical registration, payment, integration, and notification services.

13. Reporting Requirements

Authorized users shall have access to approved operational and management reports, including registration activity, section capacity/utilization, waitlist status, payment/reconciliation status, and selected audit/compliance information, subject to access controls.

14. Traceability

Traceability is maintained in the RTM. Each applicable functional requirement should map to its originating business requirement, related business rules, use case, BPMN model, NFRs, wireframe/data artifact where relevant, and test/UAT coverage.

15. Acceptance & Verification

Requirements are considered ready for business acceptance when they are approved, implemented, traceable to applicable test/UAT scenarios, and the agreed acceptance criteria are satisfied without unresolved critical defects.

Verification Method	Used For
Inspection / Review	Requirement wording, business rules, UI/data consistency
Functional Testing	Functional system behavior
Integration Testing	External-system exchanges and failure handling
Performance / Concurrency Testing	Response times, peak load, seat concurrency
Security Testing	Authentication, authorization, session and data protection
UAT	Business-user validation of end-to-end business outcomes

Appendix A – Recommended SRS Artifact References

Artifact	Purpose
----------	---------

Functional Requirements Catalog	Detailed FR-001 to FR-035 catalog
Non-Functional Requirements Catalog	Detailed measurable NFRs
Use Case Specification Master	Detailed actor/system flows
Business Rules Catalog.xlsx	Authoritative business-rule definitions
BPMN Models	Process flows, decisions and responsibilities
Wireframes	UI behavior and screen structure
Data Model & Data Dictionary	Logical entities, attributes and relationships
RTM.xlsx	End-to-end requirement traceability
UAT Plan & Scenarios	Business acceptance coverage

Appendix B – Quality Checklist

- ☐ Every FR has a unique ID and testable requirement statement.
- ☐ Every important FR is linked to its Use Case .
- ☐ NFRs are measurable or have a defined verification method.
- ☐ Business rules are referenced consistently and not duplicated inconsistently.
- ☐ Interfaces identify authoritative outcomes and failure handling.
- ☐ Data and UI requirements reference the detailed portfolio artifacts.
- ☐ RTM provides end-to-end coverage through test/